Audio Formats Comparison

Format	Type	Encoding	Pros	Cons	Common Use Case
WAV	Uncompressed	PCM (raw samples)	Sounds exactly like the original recording; easy to edit without losing quality	Takes up a lot of storage space (~10MB per minute); doesn't have a standard way to store song info	Studio recording, editing, mastering
AIFF	Uncompressed	PCM (Apple's WAV equivalent)	Sounds exactly like the original recording; can store more song info (like title, artist)	Just as large as WAV files; works best only on Apple/Mac devices	Mac-based audio production
MP3	Lossy compressed	Psychoacoustic perceptual coding	Very small files that play on virtually any device or app	Loses some sound quality forever; can sound rough at low quality settings; older technology compared to newer small-file formats	Streaming, general listening, podcasts
FLAC	Lossless compressed	Linear prediction + Rice coding	Sounds identical to the original but takes up less space; free to use, no licensing fees	Bigger files than MP3 and similar formats; not every device or player supports it	Archiving, audiophile distribution
ALAC	Lossless compressed	Apple's lossless codec (similar approach to FLAC)	Sounds identical to the original; works smoothly with iPhones, iTunes, and other Apple apps	Not as well supported outside of Apple devices and apps	Apple Music lossless tier, iTunes libraries

Grouping Summary

- Uncompressed (WAV, AIFF) → reference quality, editing/production
- Lossy compressed (MP3) → small size, quality loss, general playback/streaming
- Lossless compressed (FLAC, ALAC) → smaller than uncompressed, no quality loss, archiving/high-fidelity listening

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